

ADITYA KULKARNI

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EDUCATION

North Carolina State University

M.S. Computer Science

Raleigh, NC

Aug 2024 – May 2026

University of Mumbai

B.Tech. Computer Science, CGPA: 9.32/10

Mumbai, India

Aug 2020 – May 2024

WORK EXPERIENCE

Product Engineer Intern

May 2025 – Present

Isomer AI

Remote

- Shipped 10+ production backend features for an AI-powered claims automation platform – multi-model LLM tooling across 3 providers (Claude, GPT-4o, Gemini) with MCP, Temporal.io workflow orchestration, and RBAC – across 7 deployments using TypeScript, Node.js, PostgreSQL, and Redis
- Built document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage) and real-time event pipelines using Pusher; contributed 394 commits and led code review across high-velocity Agile cycles

Software Developer Intern

Jun 2024 – Aug 2024

Mphasis (HP Company)

Remote

- Built a health claim automation pipeline using OCR, NLP, and ML models to extract and validate structured data from unstructured documents, improving claim processing accuracy by 25% in production
- Automated CI/CD deployments on AWS and GCP; integrated Grafana monitoring dashboards and conducted model validation and red-teaming to improve production reliability

Graduate Research Assistant

Apr 2025 – Present

NC State University

Raleigh, NC

- Engineered a cloud-based genomic data platform on Google Cloud Run with JBrowse 2, enabling interactive visualization of 500GB datasets for 100+ researchers worldwide
- Improved ML model accuracy 12% through hyperparameter tuning and time-series analysis; collaborated on architecture design and model evaluation for biological datasets

PERSONAL PROJECTS

Glance AI – Privacy-First RAG System | Python, LLMs, Qdrant, LangChain, FastAPI

2025

- Built a compliance-ready RAG system for fintech and healthcare with per-tenant vector retrieval isolation, preventing cross-tenant context leakage and LLM hallucinations for regulated data environments
- Designed deterministic output architecture ensuring factual correctness for sensitive financial and medical records; validated against privacy constraints modeled after SOC 2 and HIPAA data boundaries

LLM Eval Harness | Python, Anthropic/OpenAI/Gemini SDKs, DuckDB, Streamlit, Docker

2025

- Built a vendor-neutral LLM benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across grounded QA, multi-hop QA, and claim verification on a 2,000-example frozen test set with 3 repeated runs per model
- Tracked ExactMatch, TokenF1, hallucination rate, p50/p95 latency, and cost/call; applied paired bootstrap CIs, McNemar test, and Holm correction – enabling statistically rigorous model selection for production AI systems

KubeServe – Kubernetes LLM Inference Platform | Python, FastAPI, Redis, Kubernetes, KEDA

2026

- Built a Kubernetes-native LLM inference platform: FastAPI router with 3-state circuit breaker and EWMA latency-aware load balancing across Claude, GPT-4o, and Gemini; KEDA autoscaler scales model pods 0→N on queue depth
- Python Kopf CRD operator reconciles Deployments and KEDA ScaledObjects; OpenTelemetry + Prometheus + Grafana observability; k6 load-tested to 500 VUs; 88 pytest tests; Helm + ArgoCD GitOps

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, JavaScript, R, SQL, C

AI/ML: PyTorch, TensorFlow, LangChain, HuggingFace, OpenAI API, RAG, NLP, OCR, Pandas, NumPy

Infrastructure: AWS, GCP, Docker, Kubernetes, Kafka, Prometheus, Grafana, Terraform, CI/CD

Frameworks & Tools: FastAPI, Node.js, Temporal.io, PostgreSQL, Redis, Spring Boot